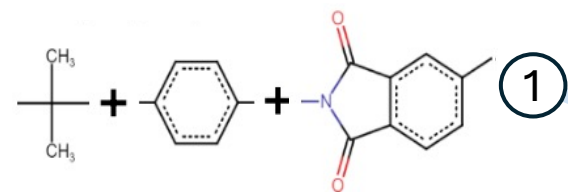


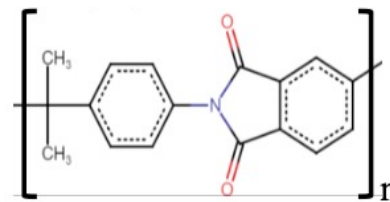
Physical-based Synthetic Data Generation

Polymer Chemical Language Model

Sample Groups



Generate Polymers



P-SMILES

Nc1ccc2oc(C()C)nc2c1

Tokenize
Tokens
 $\mathbf{p} \in \mathbb{R}^{202}$

Tokenization Layer

Token Embeddings

Asterisk Encoder

$f_A(\cdot; \theta_A)$

Asterisk Embeddings

Backbone Encoder

12 Transformer Blocks
 $f_{\text{backbone}}(\cdot; \hat{\theta}_B)$

Initial Token Embeddings

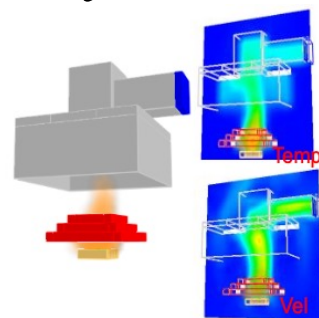
Token Embeddings
 $\mathbf{x} \in \mathbb{R}^{202 \times 768}$

Correlate Physical Quantities

$$P = \frac{\sum_i N_i p_i}{\sum_i N_i M_i}$$

P : a polymer property
 p_i : group- i contribution to P
 M_i : group- i molar mass
 N_i : group- i occurrence

Fire Dynamics Simulator



Fire Properties

Predictor
 $r(\cdot; \xi)$

Decoder Module

Predicted pSMILES

Predicted Tokens
 $\hat{\mathbf{p}} \in \mathbb{R}^{202}$

Classifier

$g(\cdot; \psi)$

Reconstructed Token Embedding
 $\hat{\mathbf{x}} \in \mathbb{R}^{202 \times 768}$

Decoder Layer

$h(\cdot; \phi)$

Aggregation
pSMILES Embeddings
 $\mathbf{z} \in \mathbb{R}^{768}$